

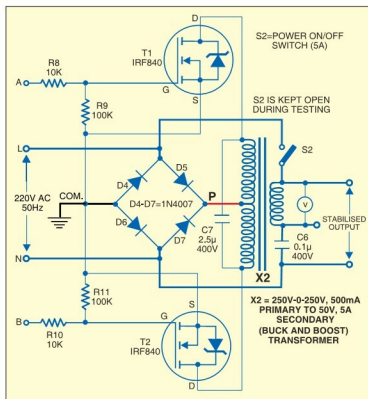


Fig. 2: Circuit of voltage stabiliser buck-boost section

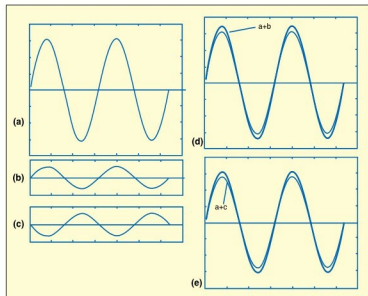


Fig. 3: (a) AC mains voltage waveforms, (b) small voltage of about 30V in phase with 'a', (c) a small voltage of about a 30V out of phase with 'a', (d) waveforms with 'a+b', (e) waveforms with 'a+c'

pins. So at points marked 'A' and 'B' we get two pulse trains from the microcontroller. Synchronisation with the mains supply is achieved by the square wave (50Hz mains derived) on port-D bit 7 (pin 30).

Transistor T3 (BC547) produces a rectangular pulse from the half-wave-rectified low voltage (9V) from the transformer (9V-0-9V, 300mA). Using

50 Hz as reference for positive and negative half cycles of the mains supply, it produces the pulses at A and B points in turn. These pulses change in width and are hence called pulse-width-modulated. The width varies in accordance with the voltage to be produced for compensating the voltage from mains supply.

After wiring the circuit, program the chip with the given Assembly program. Insert the chip into the board and apply power supply. The chip has two PWM pins, 16 and 17. Adjust the shaft of potmeter VR1 (10-kilo-ohm) to the bottom position for zero voltage. Also, ground pin D. The PWM pulse is now available from pin 17 of IC2, while pin 16 is low. If the shaft of the potmeter is moved to the top position when 'D' is connected to ground, pulses will be available from pin 16.

Taking pin D to 5V reverses the above sequence.

After checking this part of the circuit, the circuit shown in Fig. 2 may be tested.

In manual position of input-selection switch S1 (Fig. 1), the analogue input voltage from potmeter

PARTS LIST

Semiconductors:

- IC1 - 7805 voltage regulator
- IC2 - PIC16F877A microcontroller
- IC3 - ULN2003 buffer
- D1-D8 - 1N4007 diode
- T1, T2 - IRF840 MOSFET
- T3 - BC547 npn bipolar transistor
- ZD1 - 10V zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R3-R8, - 10-kilo-ohm
- R10, R12, R13 - 10-kilo-ohm
- R2 - 4.7-kilo-ohm
- R9, R11 - 100-kilo-ohm
- VR1 - 10-kilo-ohm potmeter
- VR2 - 5-kilo-ohm preset

Capacitors:

- C1 - 100nF ceramic
- C2 - 1000µF, 25V electrolytic
- C3, C4 - 22pF ceramic
- C5 - 10µF, 25V electrolytic
- C6 - 0.1µF, 400V AC
- C7 - 2.5µF, 400V AC

Miscellaneous:

- X1 - 230V AC primary to 9V-0-9V, 300mA secondary transformer
- X2 - 250V-0-250V, 500mA primary to 50V, 5A secondary transformer
- X_{VAL} - 4MHz crystal oscillator
- S1 - SPDT changeover switch
- S2 - On/off power switch (5A)

VR1 is used. In this position, the circuit functions as a variac that varies the output voltage from 180V to 250V as the potmeter is varied.

In auto position of S1, the circuit acts as a stabiliser. For this, transformer-rectified supply derived from the mains provides a proportional voltage to the ADC of the chip.

Point E in the voltage stabiliser controller circuit gives a voltage that varies with mains voltage. At exactly 220V mains, the 9V transformer (X1) gives a peak voltage of $9\sqrt{2} = 12.7V$ and subtracting 10V using zener diode ZD1 gives 2.7V at point E. It increases to 5V when the mains voltage rises to 250V and drops to zero when mains drops to 172V – in effect, giving 0 to 5V over this range.

Using this voltage at point E, you can assess variation in the mains voltage and thereby control the PWM-based sine voltage for adding (boost) or subtracting (buck) from mains. Point E is connected to the ADC input pin